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جامعة الأقصى

كلية الحاسبات وتكنولوجيا المعلومات

قسم الهندسة والشبكات

## **Network Project**

### **IP Addressing and Subnet Mask Calculation**

**Final Project – Comouter Networks**

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## 1. Project Addressing Idea

In this final project, I used two types of IP addressing. For the floor networks and servers, I used Class C networks with /24 because these networks contain PCs, laptops, servers, and access points. For router-to-router links, I used /30 because each link needs only two usable IP addresses, one for each router.

## 2. Basic Binary Table Used in the Calculation

The calculation depends on the normal 8-bit table below. This table helps convert the subnet mask from binary to decimal.

| 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 |
|-----|----|----|----|---|---|---|---|
| 1   | 1  | 1  | 1  | 1 | 1 | 1 | 1 |
| 1   | 1  | 1  | 1  | 1 | 1 | 0 | 0 |

Example:  $255 = 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1$ , so 255 in binary is 11111111.

Example:  $252 = 128 + 64 + 32 + 16 + 8 + 4$ , so 252 in binary is 11111100.

## 3. Main Formulas

| Item           | Formula / Rule                                                                       |
|----------------|--------------------------------------------------------------------------------------|
| Host bits      | Host Bits = $32 - \text{Prefix}$                                                     |
| Usable hosts   | Usable Hosts = $2^{(\text{Host Bits})} - 2$                                          |
| Block size     | Block Size = $256 - \text{value of the subnet mask in the changing octet}$           |
| Classful rule  | If the subnet mask is the default mask of the class, it is Classful.                 |
| Classless rule | If the subnet mask is different from the default mask of the class, it is Classless. |

## 4. IP Classes Used in the Project

| Class   | First Octet Range | Default Mask  | Prefix | Example in Project |
|---------|-------------------|---------------|--------|--------------------|
| Class A | 1 - 126           | 255.0.0.0     | /8     | 10.0.0.0           |
| Class B | 128 - 191         | 255.255.0.0   | /16    | 172.16.1.0         |
| Class C | 192 - 223         | 255.255.255.0 | /24    | 192.168.4.0        |

For example, 192.168.4.0 is Class C because the first octet is 192. The default mask of Class C is 255.255.255.0, so 192.168.4.0/24 is Classful.

## 5. Floor Networks Calculation (/24)

I used /24 for the floor networks because each floor can contain several devices. The /24 network gives enough addresses and is simple to manage.

### Example: 192.168.4.0/24

This network was used for the Administration Building server room. The gateway is 192.168.4.1 and the servers are in the same network.

Mask calculation:

255.255.255.0 = 11111111.11111111.11111111.00000000

Prefix = /24, so Network Bits = 24

Host Bits = 32 - 24 = 8

Usable Hosts =  $2^8 - 2 = 256 - 2 = 254$  devices

| Item              | Value         |
|-------------------|---------------|
| Network Address   | 192.168.4.0   |
| First Usable IP   | 192.168.4.1   |
| Last Usable IP    | 192.168.4.254 |
| Broadcast Address | 192.168.4.255 |
| Subnet Mask       | 255.255.255.0 |
| Number of Devices | 254           |
| Type              | Classful      |

### Floor Networks Used in the Project

| Network         | Used For                          | Gateway      |
|-----------------|-----------------------------------|--------------|
| 192.168.4.0/24  | Administration Floor 1 / Servers  | 192.168.4.1  |
| 192.168.5.0/24  | Administration Floor 2 / Wireless | 192.168.5.1  |
| 192.168.10.0/24 | Library Floor 1 / Wired           | 192.168.10.1 |
| 192.168.11.0/24 | Library Floor 2 / Wireless        | 192.168.11.1 |
| 192.168.20.0/24 | Engineering Floor 1 / Wired       | 192.168.20.1 |
| 192.168.21.0/24 | Engineering Floor 2 / Wireless    | 192.168.21.1 |
| 192.168.30.0/24 | Business Floor 1 / Wired          | 192.168.30.1 |
| 192.168.31.0/24 | Business Floor 2 / Wireless       | 192.168.31.1 |
| 192.168.40.0/24 | Health Floor 1 / Wired            | 192.168.40.1 |
| 192.168.41.0/24 | Health Floor 2 / Wireless         | 192.168.41.1 |

All floor networks use mask 255.255.255.0. The calculation for each one is the same: Host Bits = 8 and Usable Hosts = 254. These networks are Classful because they are Class C with the default /24 mask.

## 6. Mesh Router Links Calculation (/30)

For the mesh links between the five main building routers, I used 10.0.0.x/30. This is suitable because each link connects only two routers.

### Example: 10.0.0.36/30

This network was used for the link between BUSECO-RTR and HEALTHSCI-RTR.

10.0.0.36 is Class A, and the default Class A mask is 255.0.0.0. But I used 255.255.255.252, so this is Classless.

Mask calculation:

252 = 128 + 64 + 32 + 16 + 8 + 4 = 11111100

255.255.255.252 = 11111111.11111111.11111111.11111100

Host Bits = 32 - 30 = 2

Usable Hosts =  $2^2 - 2 = 2$  devices

Block Size = 256 - 252 = 4

| Item            | Value     |
|-----------------|-----------|
| Network Address | 10.0.0.36 |

| First Usable IP     |              | 10.0.0.37       |                  |
|---------------------|--------------|-----------------|------------------|
| Second Usable IP    |              | 10.0.0.38       |                  |
| Broadcast Address   |              | 10.0.0.39       |                  |
| Subnet Mask         |              | 255.255.255.252 |                  |
| Number of Devices   |              | 2               |                  |
| Type                |              | Classless       |                  |
| Link                | Network      | First Router IP | Second Router IP |
| ADMIN - LIBRARY     | 10.0.0.0/30  | 10.0.0.1        | 10.0.0.2         |
| ADMIN - ENGTECH     | 10.0.0.4/30  | 10.0.0.5        | 10.0.0.6         |
| ADMIN - BUSECO      | 10.0.0.8/30  | 10.0.0.9        | 10.0.0.10        |
| ADMIN - HEALTHSCI   | 10.0.0.12/30 | 10.0.0.13       | 10.0.0.14        |
| LIBRARY - ENGTECH   | 10.0.0.16/30 | 10.0.0.17       | 10.0.0.18        |
| LIBRARY - BUSECO    | 10.0.0.20/30 | 10.0.0.21       | 10.0.0.22        |
| LIBRARY - HEALTHSCI | 10.0.0.24/30 | 10.0.0.25       | 10.0.0.26        |
| ENGTECH - BUSECO    | 10.0.0.28/30 | 10.0.0.29       | 10.0.0.30        |
| ENGTECH - HEALTHSCI | 10.0.0.32/30 | 10.0.0.33       | 10.0.0.34        |
| BUSECO - HEALTHSCI  | 10.0.0.36/30 | 10.0.0.37       | 10.0.0.38        |

All mesh links use mask 255.255.255.252. Each network has 2 usable IP addresses, which is enough for a point-to-point router link.

## 7. Building Router to Floor Router Links (/30)

I also used /30 for the links between the main building router and the floor routers. This is the same idea: each link has only two router interfaces.

### Example: 172.16.2.0/30

This network was used for the link between LIBRARY-RTR and LIB-F1-RTR.

172.16.2.0 is Class B, and the default Class B mask is 255.255.0.0. But I used 255.255.255.252, so this is Classless.

Host Bits =  $32 - 30 = 2$

Usable Hosts =  $2^2 - 2 = 2$  devices

Block Size =  $256 - 252 = 4$

| Item              | Value           |
|-------------------|-----------------|
| Network Address   | 172.16.2.0      |
| First Usable IP   | 172.16.2.1      |
| Second Usable IP  | 172.16.2.2      |
| Broadcast Address | 172.16.2.3      |
| Subnet Mask       | 255.255.255.252 |
| Number of Devices | 2               |
| Type              | Classless       |

| Link                          | Network       | Main Router IP | Floor Router IP |
|-------------------------------|---------------|----------------|-----------------|
| ADMIN-RTR - ADMIN-F1-RTR      | 172.16.1.0/30 | 172.16.1.1     | 172.16.1.2      |
| ADMIN-RTR - ADMIN-F2-RTR      | 172.16.1.4/30 | 172.16.1.5     | 172.16.1.6      |
| LIBRARY-RTR - LIB-F1-RTR      | 172.16.2.0/30 | 172.16.2.1     | 172.16.2.2      |
| LIBRARY-RTR - LIB-F2-RTR      | 172.16.2.4/30 | 172.16.2.5     | 172.16.2.6      |
| ENGTECH-RTR - ENG-F1-RTR      | 172.16.3.0/30 | 172.16.3.1     | 172.16.3.2      |
| ENGTECH-RTR - ENG-F2-RTR      | 172.16.3.4/30 | 172.16.3.5     | 172.16.3.6      |
| BUSECO-RTR - BUS-F1-RTR       | 172.16.4.0/30 | 172.16.4.1     | 172.16.4.2      |
| BUSECO-RTR - BUS-F2-RTR       | 172.16.4.4/30 | 172.16.4.5     | 172.16.4.6      |
| HEALTHSCI-RTR - HEALTH-F1-RTR | 172.16.5.0/30 | 172.16.5.1     | 172.16.5.2      |
| HEALTHSCI-RTR - HEALTH-F2-RTR | 172.16.5.4/30 | 172.16.5.5     | 172.16.5.6      |

## 8. Summary

| Use                               | Example        | Mask            | Hosts | Type      |
|-----------------------------------|----------------|-----------------|-------|-----------|
| Floors and servers                | 192.168.x.0/24 | 255.255.255.0   | 254   | Classful  |
| Main router mesh links            | 10.0.0.x/30    | 255.255.255.252 | 2     | Classless |
| Main router to floor router links | 172.16.x.x/30  | 255.255.255.252 | 2     | Classless |

In conclusion, I used /24 for user and server networks because it gives enough addresses for many devices. I used /30 for router links because it gives exactly two usable IP addresses, which is the best choice for point-to-point links. This also satisfies the project requirement of using both Classful and Classless addressing.

## 9. Network Construction and Connection Method

In the Packet Tracer design, I connected the network according to the project requirements. The five main building routers are connected together using Mesh topology. Inside each building, I used Star topology, where the building router is the center and the floor devices connect through a switch or access point.

### 9.1 Main Topology Used

| Part              | Connection Method                                          | Why I Used It                                                                         |
|-------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Between buildings | Main Router <-> Main Router                                | I used Mesh so every building router has a direct path to the other building routers. |
| Floor 1 wired     | PCs -> Switch -> Floor Router -> Building Router           | This is used for wired computers on the first floor.                                  |
| Floor 2 wireless  | Laptops -> Access Point -> Floor Router -> Building Router | This is used for wireless laptops on the second floor.                                |
| Server room       | Servers -> Switch -> Floor Router -> ADMIN-RTR             | The services are placed in the Administration Building server room.                   |

### 9.2 Cable Types Used

I selected the cable type based on the devices connected together. In Packet Tracer, Auto cable can also work, but the table below explains the correct manual choice.

| Connection                      | Cable Type                  | Reason                                          |
|---------------------------------|-----------------------------|-------------------------------------------------|
| PC / Server -> Switch           | Copper Straight-Through     | End devices connect to a switch.                |
| Switch -> Floor Router          | Copper Straight-Through     | A switch connects to a router interface.        |
| Floor Router -> Building Router | Copper Cross-Over or Auto   | Both sides are router interfaces.               |
| Main Router -> Main Router      | Copper Cross-Over or Serial | This is a point-to-point router link.           |
| Floor Router -> Access Point    | Copper Straight-Through     | The access point connects to the wired network. |
| Laptop -> Access Point          | Wireless                    | The laptops connect using Wi-Fi.                |

## 10. Router Interface Configuration

After connecting the cables, I configured the router interfaces. The floor networks use /24 addresses, while router links use /30 addresses. After assigning each IP address, I used no shutdown to turn on the interface.

### 10.1 Example: Floor Network Interface

For a user network such as Library Floor 1, the gateway is placed on the floor router interface connected to the switch.

| Command                               | Purpose                                           |
|---------------------------------------|---------------------------------------------------|
| interface fastEthernet 0/0            | Open the interface connected to the floor switch. |
| ip address 192.168.10.1 255.255.255.0 | Set the gateway IP for Library Floor 1.           |
| ip helper-address 192.168.4.2         | Forward DHCP requests to the DHCP Server.         |
| no shutdown                           | Turn on the interface.                            |

### 10.2 Example: Router-to-Router Interface

For router-to-router links, I used /30 because the link needs only two usable IP addresses.

| Router        | Interface IP | Mask            | Usage                 |
|---------------|--------------|-----------------|-----------------------|
| BUSECO-RTR    | 10.0.0.37    | 255.255.255.252 | Link to HEALTHSCI-RTR |
| HEALTHSCI-RTR | 10.0.0.38    | 255.255.255.252 | Link to BUSECO-RTR    |

## 11. Static Routing Configuration

I used static routing as required in the project. The idea is that every router must know how to reach the networks that are not directly connected to it. For floor routers, I used a default route to the building router. For building routers, I added routes to the local floor networks and remote building networks.

### 11.1 Route from Floor Router to Building Router

On a floor router, I can use a default route because any network outside the floor should go to the main building router.

| Example Command                     | Explanation                                               |
|-------------------------------------|-----------------------------------------------------------|
| ip route 0.0.0.0 0.0.0.0 172.16.2.1 | Send all unknown networks from LIB-F1-RTR to LIBRARY-RTR. |

### 11.2 Route from Building Router to Floor Network

On the building router, I added routes back to its floor networks. This is important so replies can return to the correct floor.

| Example Command                                | Explanation                                             |
|------------------------------------------------|---------------------------------------------------------|
| ip route 192.168.10.0 255.255.255.0 172.16.2.2 | LIBRARY-RTR reaches Library Floor 1 through LIB-F1-RTR. |
| ip route 192.168.11.0 255.255.255.0 172.16.2.6 | LIBRARY-RTR reaches Library Floor 2 through LIB-F2-RTR. |

### 11.3 Route Between Buildings

Between the main building routers, I used the /30 mesh links as next-hop paths. For example, if ADMIN-RTR needs to reach Library networks, the next hop can be the Library router IP on the direct mesh link.

| Example Command                              | Explanation                                            |
|----------------------------------------------|--------------------------------------------------------|
| ip route 192.168.10.0 255.255.255.0 10.0.0.2 | ADMIN-RTR reaches Library Floor 1 through LIBRARY-RTR. |
| ip route 192.168.11.0 255.255.255.0 10.0.0.2 | ADMIN-RTR reaches Library Floor 2 through LIBRARY-RTR. |

## 12. DHCP and DNS Configuration

The DHCP Server is placed in the Administration Building server room. In my addressing plan, the DHCP Server IP is 192.168.4.2 and the DNS Server IP is 192.168.4.3. I created one DHCP pool for each floor network.

### 12.1 DHCP Working Idea

- The client sends a DHCP request when it is set to automatic IP addressing.
- If the client is in another network, the floor router forwards the request using ip helper-address 192.168.4.2.
- The DHCP Server gives the device an IP address from the correct pool.
- The DNS value in each pool is 192.168.4.3 so clients can resolve domain names.

### 12.2 DHCP Pools Used

| Pool      | Gateway      | DNS         | Start IP      | Mask          |
|-----------|--------------|-------------|---------------|---------------|
| ADMIN-F2  | 192.168.5.1  | 192.168.4.3 | 192.168.5.10  | 255.255.255.0 |
| LIB-F1    | 192.168.10.1 | 192.168.4.3 | 192.168.10.10 | 255.255.255.0 |
| LIB-F2    | 192.168.11.1 | 192.168.4.3 | 192.168.11.10 | 255.255.255.0 |
| ENG-F1    | 192.168.20.1 | 192.168.4.3 | 192.168.20.10 | 255.255.255.0 |
| ENG-F2    | 192.168.21.1 | 192.168.4.3 | 192.168.21.10 | 255.255.255.0 |
| BUS-F1    | 192.168.30.1 | 192.168.4.3 | 192.168.30.10 | 255.255.255.0 |
| BUS-F2    | 192.168.31.1 | 192.168.4.3 | 192.168.31.10 | 255.255.255.0 |
| HEALTH-F1 | 192.168.40.1 | 192.168.4.3 | 192.168.40.10 | 255.255.255.0 |
| HEALTH-F2 | 192.168.41.1 | 192.168.4.3 | 192.168.41.10 | 255.255.255.0 |

### 12.3 DNS Records

The DNS Server maps the service names to the IP addresses of the servers. This allows users to open services by domain name instead of using IP addresses.

| Service      | Domain Name      | Server IP   |
|--------------|------------------|-------------|
| Web Server   | www.aqsa.edu.ps  | 192.168.4.6 |
| Email Server | mail.aqsa.edu.ps | 192.168.4.5 |
| LMS Server   | lms.aqsa.edu.ps  | 192.168.4.4 |

### 13. Wireless and Security Configuration

For wireless floors, I added an access point and connected the laptops to it. The normal wireless floors use WPA2-PSK / WPA2 Personal. The Administration Floor 2 uses WPA2 Enterprise according to the project requirement.

| Network                | SSID            | Security        | Password |
|------------------------|-----------------|-----------------|----------|
| Administration Floor 2 | Admin-wif       | WPA2 Enterprise | 12345678 |
| Library Floor 2        | Library-wif     | WPA2 Personal   | 12345678 |
| Engineering Floor 2    | Engineering_wif | WPA2 Personal   | 12345678 |
| Business Floor 2       | Business_wifi   | WPA2 Personal   | 12345678 |
| Health Floor 2         | Health_wifi     | WPA2 Personal   | 12345678 |

All wireless networks were tested using the password 12345678. After the laptop connects to the correct SSID, it uses DHCP to receive the IP address, subnet mask, gateway, and DNS from the central DHCP server.

### 14. Testing Plan

After finishing the connections, IP addressing, DHCP, DNS, and routing, I tested the network step by step. This makes it easier to find errors.

| Test                | How I Tested It                   | Expected Result                                  |
|---------------------|-----------------------------------|--------------------------------------------------|
| DHCP                | Set PCs and laptops to DHCP       | Each device receives IP, mask, gateway, and DNS. |
| Gateway Ping        | Ping the floor gateway            | Device can reach its router.                     |
| Inter-building Ping | Ping a device in another building | Static routing works.                            |
| DNS                 | Ping www.aqsa.edu.ps              | Name resolves to the Web Server.                 |
| Web                 | Open www.aqsa.edu.ps in browser   | University web page opens.                       |
| Email               | Send email between two users      | Email service works.                             |

This implementation follows the project idea: mesh links between the main building routers, star topology inside each building, DHCP and DNS services from the Administration server room, static routing between subnets, and secured wireless networks.